

Exercise Sheet 10b - Solutions

Predicate Logic – Natural Deduction & Semantics

1. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow \neg \exists x. \forall y. \neg x < y$

$$\begin{array}{c}
 \frac{\frac{\frac{\frac{\overline{S_1}^1}{\exists y. x < y}^4 [\forall E] \quad \frac{\frac{\overline{x < y}^4 \quad \frac{\overline{\forall y. \neg x < y}^3}{\neg x < y} [\forall E]}{\perp}^4 [\neg E]}{\perp}^4 [\exists E]}{\exists x. \forall y. \neg x < y}^2 \quad \perp^3 [\exists E]}{\perp}^2 [\neg I]}{\neg \exists x. \forall y. \neg x < y}^1 [\rightarrow I]} \\
 \frac{\perp^2 [\neg I]}{\neg \exists x. \forall y. \neg x < y}^2 [\neg I]} \\
 \frac{\neg \exists x. \forall y. \neg x < y}^{(S_1) \rightarrow \neg \exists x. \forall y. \neg x < y}^1 [\rightarrow I]}
 \end{array}$$

2. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow S_3$

$$\begin{array}{c}
 \frac{\frac{\overline{S_1}^1}{\exists y. 0 < y}^1 [\forall E] \quad \frac{\frac{\frac{\overline{S_2}^2}{\forall x. 0 < y \rightarrow 1 \leq y}^2 [\forall E] \quad \frac{\overline{0 < y \rightarrow 1 \leq y}^3 [\forall E]}{0 < y \rightarrow 1 \leq y}^3 [\forall E]}{1 \leq y}^3 [\rightarrow E]}{\frac{1 \leq y}{S_3}^3 [\exists I]} \\
 \frac{\frac{\overline{S_1}^1}{\exists y. 0 < y}^1 [\forall E] \quad \frac{1 \leq y}{S_3}^3 [\exists I]}{S_3}^3 [\exists E]} \\
 \frac{S_3}{(S_2) \rightarrow S_3}^2 [\rightarrow I]} \\
 \frac{(S_2) \rightarrow S_3}{(S_1) \rightarrow (S_2) \rightarrow S_3}^1 [\rightarrow I]}
 \end{array}$$

3. For example, the models

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n \leq m \} \rangle \rangle$
- and $M'_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n \leq m \}, \emptyset \rangle \rangle$
- and $M''_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid \text{True} \}, \emptyset \rangle \rangle$

are models of S_1 ;

4. For example, the models

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n > m \}, \emptyset \rangle \rangle$
- and $M'_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \emptyset \rangle \rangle$
- and $M''_2 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \emptyset \rangle \rangle$

are models of $\neg S_1$.